Sales Analytics: Insights Report

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Introduction

This project analyses the sales data of a small online store using PostgreSQL. The database has three tables: customers (customer details and city), products (product name, category and price) and orders (what each customer bought, with quantity, amount and date). Using SQL queries, including joins, grouping, subqueries, CTEs and CASE statements, the project answers business questions about monthly revenue, revenue by city, top product categories, repeat customers and customer segments. Two views, monthly sales view and customer summary view, were also created to make this reporting reusable. The key findings are summarised as five insights in this report.

Dataset Overview

The dataset contains 12 customers across 10 cities, 8 products in 4 categories, and 24 orders placed between January and September 2024. Total revenue is 360,100, with an average order value of about 15,004. Ten of the 12 customers placed at least one order. The tables are linked by foreign keys: each order refers to one customer and one product.

Key Insights

Insight 1: Revenue peaks in July and is lowest in January

Finding: July 2024 earned the highest revenue (68,750), while January earned the lowest (3,750).
Evidence: Query Q2 (monthly revenue).
Meaning: Revenue is driven by a few high-priced electronics orders, such as laptops and smartphones, not by the number of orders. Months with a laptop or smartphone order earn far more than months with only small items.
Recommendation: Promote high value electronics in slow months, and plan stock for the months where those orders cluster.

Insight 2: Kochi earns the most, and some cities have customers but no revenue

Finding: Kochi earns the most revenue (119,100), followed closely by Bangalore (115,000), which has only one customer. Kannur and Palakkad have registered customers but zero revenue.
Evidence: Queries Q4 (revenue by city) and Q18 (customers and revenue per city).
Meaning: A few cities drive most of the revenue, and Kochi and Bangalore together account for about 65% of total revenue (234,100 of 360,100). Revenue depends on high-value purchases like laptops and smartphones, not on the number of customers. Cities with sign ups but no orders show customers who haven't been converted.
Recommendation: Run a first order offer for customers in Kannur and Palakkad, and look at what Bangalore's customer buys so it can be promoted to similar customers elsewhere.

Insight 3: Electronics leads by revenue, but Stationery leads by units

Finding: Electronics is the top category by revenue (302,500, 84% of total), while Stationery is the top by units (21 units) but earns only 1.83% of revenue.
Evidence: Queries Q6, Q7 and Q8 (category revenue, share of total, and top category by units).
Meaning: "Top selling" depends on how you measure it. Stationery drives volume, but Electronics drives the money. Laptops and smartphones create most of the revenue from a small number of orders.
Recommendation: Protect and promote Electronics, since it is the revenue engine. Use low price stationery to attract customers, and try bundling it with higher value items to raise order value.

Insight 4: Repeat customers generate almost all the revenue

Finding: 8 of the 10 customers who ordered are repeat customers, and they generate 99.1% of revenue (356,850 of 360,100), spending on average 44,606 each against 1,625 for one time customers.
Evidence: Queries Q11, Q12 and Q19 (repeat customers, repeat vs one time, and spend by type).
Meaning: The business depends on customers who come back. One-time buyers contribute very little, and two registered customers have never ordered at all.
Recommendation: Focus on retention (loyalty offers and follow up messages after the first purchase) and convert one time and non ordering customers into repeat buyers.

Insight 5: A few Premium customers drive revenue, and a third of customers add almost nothing

Finding: Only 3 of 12 customers (25%) are Premium, but they generate 65% of revenue (234,100 of 360,100). At the other end, 4 customers (33%) are either Low value (2) or Inactive (2), and together they contribute under 1% of revenue.
Evidence: Queries Q14 (customer segments) and Q20 (segment summary).
Meaning: Revenue is concentrated in a few high spending customers, so losing one Premium customer would hurt a lot. A third of the customer base adds almost nothing.
Recommendation: Protect Premium customers with loyalty perks, move Regular customers toward Premium with targeted offers, and run a first-purchase campaign for the 2 inactive and 2 low value customers.

Views Created

Two views were created to make the analysis reusable. Monthly_sales_view summarises orders, revenue and average order value for each month. customer_summary_view shows each customer's number of orders, total spend and last order date, and includes customers who have never ordered. Both can be queried like normal tables with a simple SELECT.

Conclusion

This project used SQL in PostgreSQL to analyse the sales of a small online store, using three linked tables (customers, products and orders). The analysis shows that revenue depends on a small number of high-value purchases and a small group of loyal customers.

1. Electronics earns 84% of revenue, mainly from laptops and smartphones. Stationery sells the most units but contributes under 2% of revenue.
1. Revenue is uneven. It peaks in July and is lowest in January, and Kochi and Bangalore together bring in about 65% of the total.
2. Repeat customers make up 8 of the 10 buyers and generate 99.1% of revenue.
3. Premium customers are 25% of the customer base but generate 65% of revenue, while a third of customers (inactive or low value) add almost nothing.

Recommendations

1. Retain high value customers with loyalty offers, since losing even one Premium customer would hurt revenue.
2. Convert inactive and one time customers (for example in Kannur and Palakkad) with first purchase offers and follow ups.
3. Promote Electronics in slow months, and use low priced stationery to attract customers and bundle it with higher value items.

Two views, monthly sales view and customer summary view, were created so this analysis can be refreshed and reused with a simple SELECT. The dataset is small (12 customers and 24 orders), so these findings apply to this dataset, and a larger dataset would be needed to confirm them.